MATH 345 Skeleton Notes
Unit 4: Orthogonality and least squares
Section 4.3: Least squares as projection

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Revisiting a requested output as equations (U2-LO1, U2-LO3) asked whether a target vector $\mathbf{b}$ is reachable as $A\mathbf{x}$.
Recall the matrix and target from that activity:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

For

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we have

$$A\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}.$$

Every reachable output therefore has third coordinate equal to the sum of its first two coordinates. The target $\mathbf{b}$ is not reachable because $4 \neq 1 + 2$.
More generally, when $\mathbf{b} \notin \operatorname{col}(A)$, no input reaches the target exactly. Least squares replaces the unreachable target by the closest reachable output.

Definition: Least-squares solution, fitted vector, and residual

Let A be an $m \times n$ matrix and let $\mathbf{b} \in \mathbb{R}^m$.

A **least-squares solution** is a vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ that minimizes

$$\|A\mathbf{x} - \mathbf{b}\|^2$$

over all $\mathbf{x} \in \mathbb{R}^n$.
The vector

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}}$$

is the **fitted vector**. The vector

$$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}} = \mathbf{b} - A\hat{\mathbf{x}}$$

is the **residual**.
The quantity

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^{m} \big((A\mathbf{x})_i - b_i\big)^2$$

is the sum of the squares of the coordinate-wise errors. This explains the name least squares. Minimizing the norm or its square gives the same minimizers.
If $\mathbf{b} \in \mathrm{col}(A)$, the minimum error is 0, and least squares agrees with solving $A\mathbf{x} = \mathbf{b}$ exactly. If the target is not reachable, the fitted vector $\hat{\mathbf{b}}$ is the closest reachable output.

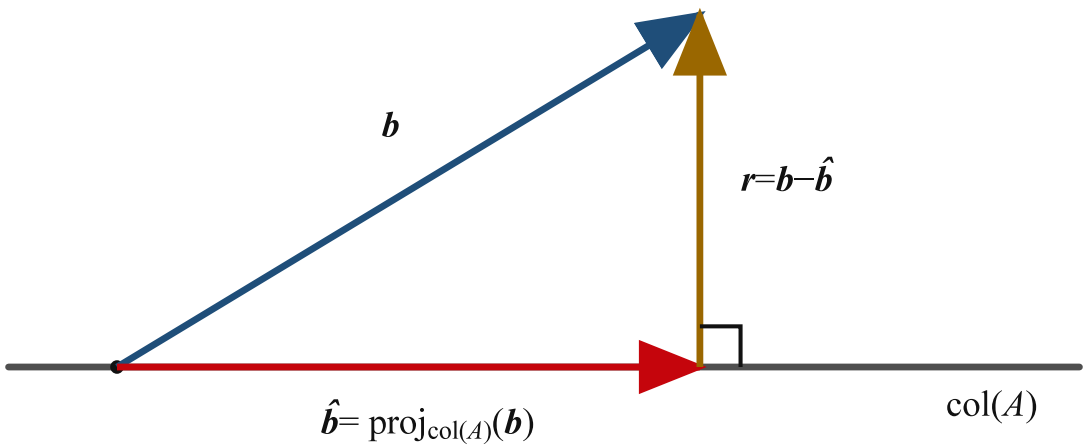


Figure. The target vector $\mathbf{b}$, its fitted vector $\widehat{\mathbf{b}}$ in $\mathrm{col}(A)$, and the perpendicular residual $\mathbf{r}$.

The residual is

$$\mathbf{r} = \mathbf{b} - \widehat{\mathbf{b}}.$$

Projection theorem guarantees a closest vector $\widehat{\mathbf{b}} \in \mathrm{col}(A)$. Since $\widehat{\mathbf{b}}$ lies in the column space, at least one coefficient vector $\widehat{\mathbf{x}}$ satisfies

$$A\widehat{\mathbf{x}} = \widehat{\mathbf{b}}.$$

Therefore the least-squares fitted vector is

$$\widehat{\mathbf{b}} = A\widehat{\mathbf{x}} = \mathrm{proj}_{\mathrm{col}(A)}(\mathbf{b}).$$

Its residual

$$\mathbf{r} = \mathbf{b} - A\widehat{\mathbf{x}}$$

is orthogonal to $\mathrm{col}(A)$.

Fact: A^T checks column orthogonality

Write

$$A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}.$$

For any vector $\mathbf{r} \in \mathbb{R}^m$,

$$A^T\mathbf{r} = \begin{bmatrix} \mathbf{a}_1^T\mathbf{r} \\ \vdots \\ \mathbf{a}_n^T\mathbf{r} \end{bmatrix}.$$

Therefore

$$A^T\mathbf{r} = \mathbf{0}$$

exactly when $\mathbf{r}$ is orthogonal to every column of A. Equivalently,

$$\mathrm{col}(A)^\perp = \mathrm{null}(A^T).$$

Theorem: Least squares and the normal equations

A vector $\widehat{\mathbf{x}}$ is a least-squares solution exactly when the residual

$$\mathbf{r} = \mathbf{b} - A\widehat{\mathbf{x}}$$

is orthogonal to $\mathrm{col}(A)$. Equivalently,

$$A^T\left(\mathbf{b} - A\widehat{\mathbf{x}}\right) = \mathbf{0}.$$

Thus the least-squares solutions are exactly the solutions of the normal equations

$$A^TA\widehat{\mathbf{x}} = A^T\mathbf{b}.$$

Why is this true?

Every candidate fitted vector $A\mathbf{x}$ lies in $\mathrm{col}(A)$. By Projection theorem, $A\widehat{\mathbf{x}}$ is closest to $\mathbf{b}$ exactly when

$$\mathbf{b} - A\widehat{\mathbf{x}} \in \mathrm{col}(A)^\perp.$$

By A^T checks column orthogonality, this is equivalent to

$$A^T\left(\mathbf{b} - A\widehat{\mathbf{x}}\right) = \mathbf{0}.$$

Expanding and rearranging gives

$$A^T\mathbf{b} - A^TA\widehat{\mathbf{x}} = \mathbf{0},$$

or

$$A^TA\widehat{\mathbf{x}} = A^T\mathbf{b}.$$

Warning: Unique fitted vector, possibly nonunique coefficients

The fitted vector $\widehat{\mathbf{b}}$ is unique because orthogonal projection onto a subspace is unique. The coefficient vector $\widehat{\mathbf{x}}$ need not be unique. For any $\mathbf{z} \in \text{null}(A)$,

$$A\left(\widehat{\mathbf{x}} + \mathbf{z}\right) = A\widehat{\mathbf{x}}.$$

When the columns of A are linearly independent, $\text{null}(A) = \{\mathbf{0}\}$ by the matrix characterization of linear independence, so the least-squares coefficient vector is unique in that case.

Fact: Independent columns and A^TA

Let A be an $m \times n$ matrix. Then A^TA is invertible if and only if the columns of A are linearly independent.

Why is this true?

If

$$A^TA\mathbf{x} = \mathbf{0},$$

then

$$0 = \mathbf{x}^TA^TA\mathbf{x} = \|A\mathbf{x}\|^2.$$

Therefore $A\mathbf{x} = \mathbf{0}$. If the columns of A are linearly independent, then

$$\text{null}(A) = \{\mathbf{0}\},$$

so $\mathbf{x} = \mathbf{0}$. Thus $\text{null}(A^T A) = \{\mathbf{0}\}$, and the square matrix $A^T A$ is invertible. Conversely, suppose $A^T A$ is invertible and

$$A\mathbf{x} = \mathbf{0}.$$

Multiplying by A^T gives

$$A^T A\mathbf{x} = \mathbf{0}.$$

Since $A^T A$ is invertible, $\mathbf{x} = \mathbf{0}$. Therefore

$$\text{null}(A) = \{\mathbf{0}\},$$

so the columns of A are linearly independent.

Example: Unit 2 target revisited (U4-LO4, U4-LO5)
Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

1. Recall why $A\mathbf{x} = \mathbf{b}$ has no exact solution.
2. Form $A^T A$ and $A^T \mathbf{b}$.
3. Solve the normal equations for $\widehat{\mathbf{x}}$.
4. Compute the fitted vector

$$\widehat{\mathbf{b}} = A\widehat{\mathbf{x}}$$

and the residual

$$\mathbf{r} = \mathbf{b} - \widehat{\mathbf{b}}.$$

5. Check that

$$A^T\mathbf{r} = \mathbf{0}.$$

6. Compute $\|\mathbf{r}\|^2$.

7. Which vector is the closest reachable output: $\hat{\mathbf{x}}$ or $\widehat{\mathbf{b}}$?

8. The residual is not zero. Explain why this does not mean least squares failed.

For

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we have

$$A\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}.$$

The first two coordinates of $A\mathbf{x} = \mathbf{b}$ would force $x_1 = 1$ and $x_2 = 2$, but then the third coordinate would be 3, not 4. Thus the target is not reachable exactly.

The normal equations use

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad A^T \mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

To solve

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \widehat{\mathbf{x}} = \begin{bmatrix} 5 \\ 6 \end{bmatrix},$$

recall the 2×2 inverse formula from Determinant and inverse of a 2×2 matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \qquad ad - bc \neq 0.$$

Here $2 \cdot 2 - 1 \cdot 1 = 3 \neq 0$, so

$$(A^T A)^{-1} = \frac{1}{2 \cdot 2 - 1 \cdot 1} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Multiplying both sides of the normal equations by this inverse gives

$$\widehat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 5 \\ 6 \end{bmatrix}$$
$$= \frac{1}{3}\begin{bmatrix} 2\cdot 5 - 6 \\ -5 + 2\cdot 6 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 4/3 \\ 7/3 \end{bmatrix}.$$

The fitted vector is

$$\widehat{\mathbf{b}} = A\widehat{\mathbf{x}} = \begin{bmatrix} 4/3 \\ 7/3 \\ 11/3 \end{bmatrix},$$

and the residual is

$$\mathbf{r} = \mathbf{b} - \widehat{\mathbf{b}} = \begin{bmatrix} -1/3 \\ -1/3 \\ 1/3 \end{bmatrix}.$$

The orthogonality check gives

$$A^T \mathbf{r} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} -1/3 \\ -1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Also,

$$\|\mathbf{r}\|^2 = \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{1}{3}.$$

The coefficient vector is $\hat{\mathbf{x}}$. The closest reachable output is the fitted vector

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}}.$$

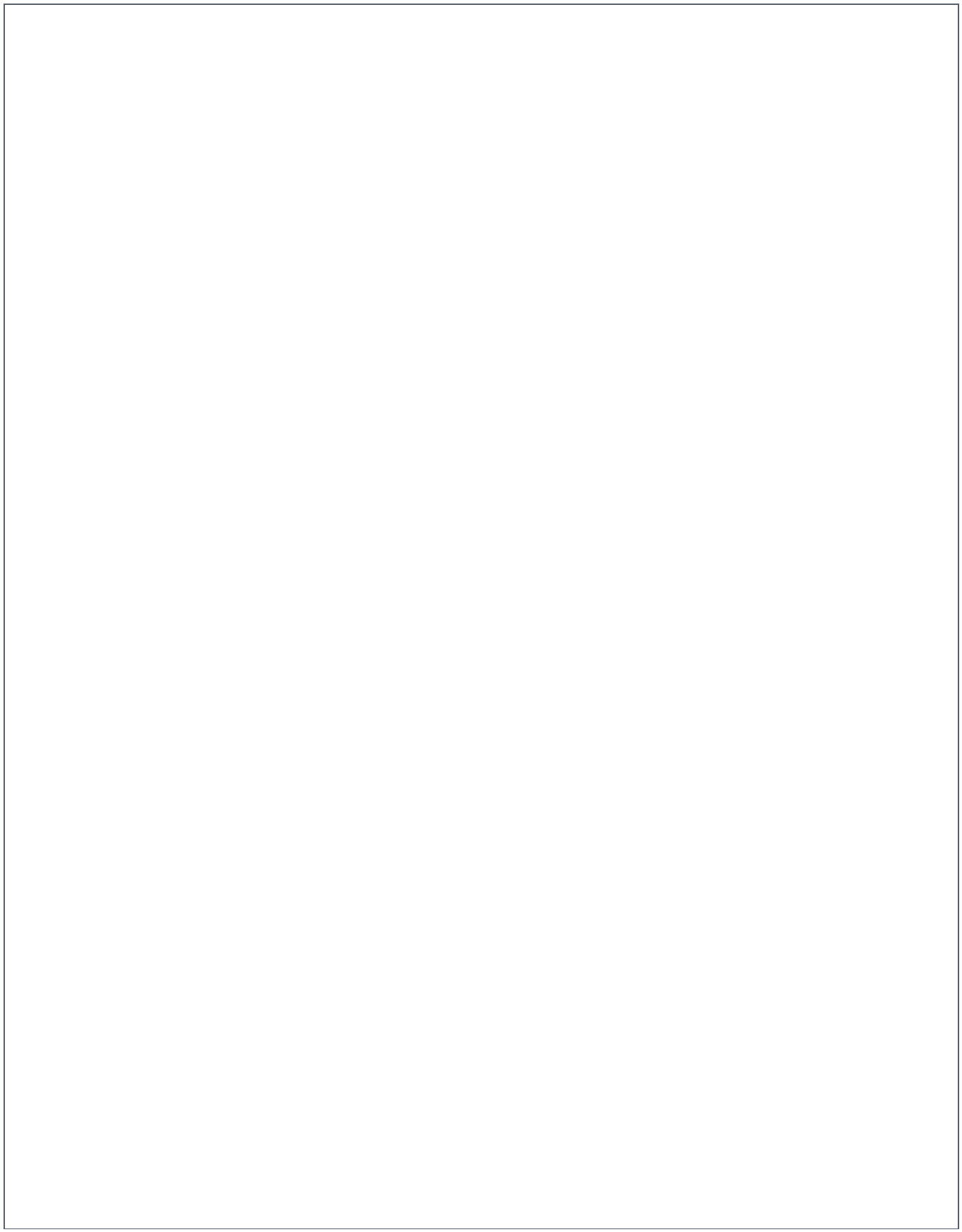
The residual is not zero because the original target is not reachable. Least squares did not fail: it found the reachable output closest to $\mathbf{b}$, and the residual is orthogonal to the column space.

Activity: Least squares and residual orthogonality (U4-LO4, U4-LO5)

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

1. Find the least-squares solution $\hat{\mathbf{x}}$ by solving the normal equations.
2. Compute the residual $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$ and check that it is orthogonal to every column of A.



Activity: Least squares with three unknowns (U4-LO4, U4-LO5)

Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}.$$

Find the least-squares solution $\widehat{\mathbf{x}}$.

Linear regression

An important practical application of least squares is

linear regression

, which fits a linear model to observed data to describe relationships and make predictions. Suppose data are given as pairs

$$(t_1, y_1), \ldots, (t_m, y_m).$$

A line model predicts

$$\hat{y} = c_0 + c_1 t.$$

The coefficient c_0 is the

intercept

, and c_1 is the

slope

.

At input t_i, the predicted value is

$$\hat{y}_i = c_0 + c_1 t_i.$$

The residual for the i-th data point is

$$r_i = y_i - \hat{y}_i.$$

Linear regression chooses c_0 and c_1 to make

$$r_1^2 + \cdots + r_m^2$$

as small as possible.
Linear regression is the same closest-reachable-output problem as Unit 2 target revisited (U4-LO4, U4-LO5), but the columns of the matrix now represent model features.

Example: A line model as a matrix product (U4-LO4)
Consider the data points

$$(0, 1), \qquad (1, 2), \qquad (2, 2),$$

and the line model

$$\hat{y} = c_0 + c_1 t.$$

1. Write the three predicted values in terms of c_0 and c_1.
2. Define

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

   Build a matrix X such that $X\mathbf{c}$ is the vector of predicted values.
3. What does the first column of X represent?

4. What does the second column of X represent?

5. Write the residual vector

$$\mathbf{r} = \mathbf{y} - X\mathbf{c}.$$

6. Explain why

$$\|\mathbf{r}\|^2 = \sum_{i=1}^{3} (y_i - \hat{y}_i)^2$$

is the sum of the squared vertical errors in the data plot.

The three predictions are

$$\hat{y}_1 = c_0, \qquad \hat{y}_2 = c_0 + c_1, \qquad \hat{y}_3 = c_0 + 2c_1.$$

Therefore

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix},$$

because

$$X\mathbf{c} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} c_0 \\ c_0 + c_1 \\ c_0 + 2c_1 \end{bmatrix}.$$

The first column is the constant feature: it records the coefficient of the intercept c_0. The second column is the t-feature: it records the input values $0, 1, 2$.
The residual vector is

$$\mathbf{r} = \mathbf{y} - X\mathbf{c} = \begin{bmatrix} 1 - c_0 \\ 2 - (c_0 + c_1) \\ 2 - (c_0 + 2c_1) \end{bmatrix}.$$

Its squared norm is

$$\|\mathbf{r}\|^2 = (1 - c_0)^2 + \big(2 - (c_0 + c_1)\big)^2 + \big(2 - (c_0 + 2c_1)\big)^2.$$

Each entry is an observed output minus the predicted output at the same input, so these are vertical errors in the data plot.

See the figure for the data points, fitted line, and vertical residuals.
For general data (t_i, y_i), define the design matrix

$$X = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \qquad \mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}.$$

Then $X\mathbf{c}$ is the vector of predicted values. Linear regression is the least-squares problem

$$\min_{\mathbf{c}} \|X\mathbf{c} - \mathbf{y}\|^2.$$

The matrix X is called the **design matrix**. Each column represents one feature used by the model.

Warning: Two different geometric pictures

The projection occurs in $\mathbb{R}^m$:

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}} = \text{proj}_{\text{col}(X)}(\mathbf{y}).$$

In the usual t-y data plot, the residuals are vertical differences

$$r_i = y_i - \hat{y}_i.$$

They are not generally perpendicular distances from the plotted points to the fitted line.

Activity: Fit a line to three data points (U4-LO4, U4-LO5)

Continue with the design matrix and data vector from A line model as a matrix product (U4-LO4). Recall that the data points are

$$(0, 1), \qquad (1, 2), \qquad (2, 2).$$

1. Use Least squares and the normal equations to form

$$X^T X\hat{\mathbf{c}} = X^T\mathbf{y}.$$

Activity: Fit a line to three data points (U4-LO4, U4-LO5) (continued)

2. Solve for

$$\hat{\mathbf{c}} = \begin{bmatrix} \hat{c}_0 \\ \hat{c}_1 \end{bmatrix}.$$

3. Write the fitted line.

4. Compute the fitted-value vector

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}}$$

   and the residual vector

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}.$$

5. Check that

$$X^T\mathbf{r} = \mathbf{0}.$$

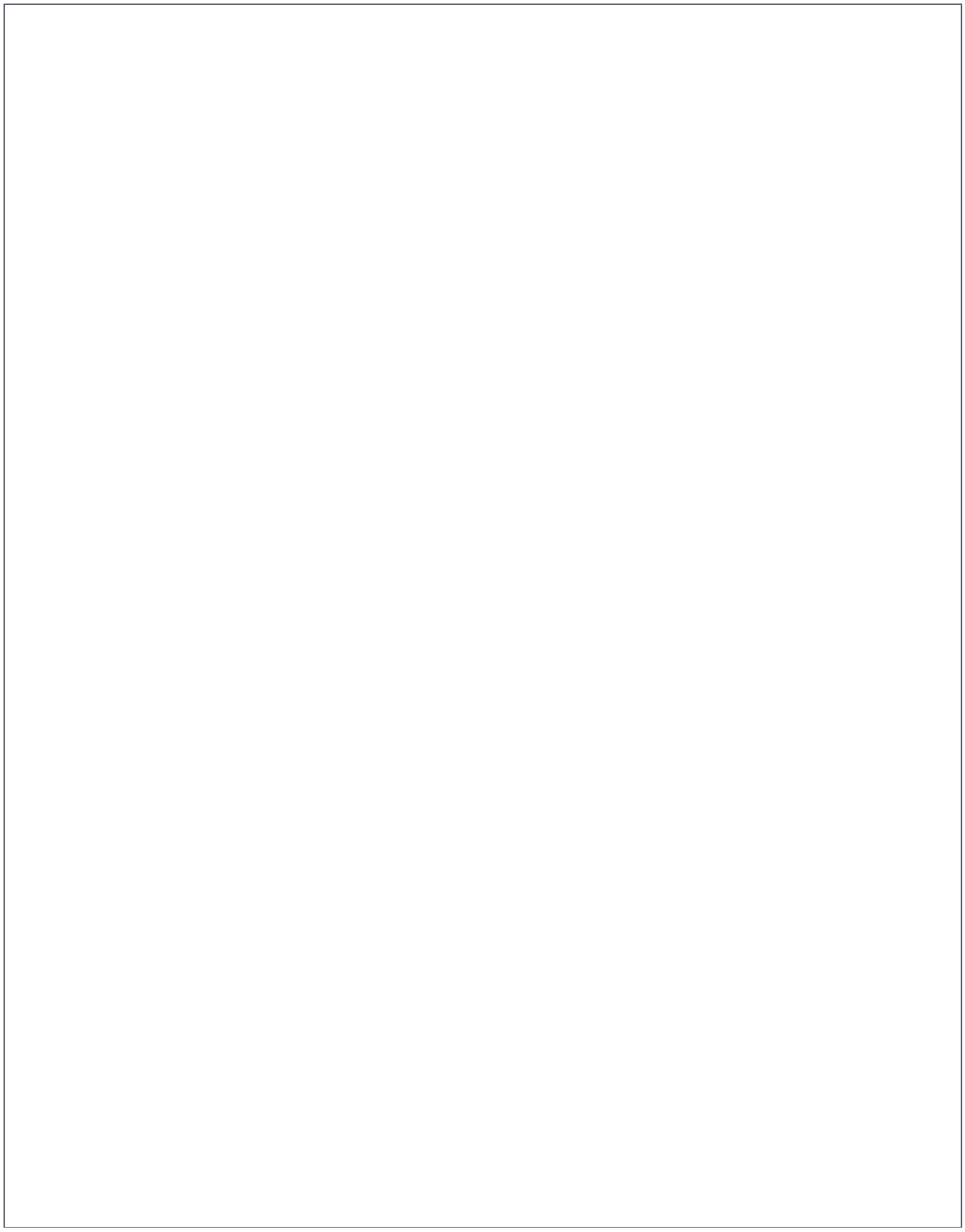
6. Write the two scalar equations contained in $X^T\mathbf{r} = \mathbf{0}$.

7. Compute

$$\|\mathbf{r}\|^2.$$

8. Which object is the fitted vector: $\hat{\mathbf{c}}$ or $\hat{\mathbf{y}}$?

9. In the t-y plot, are the residual segments vertical or perpendicular to the fitted line?



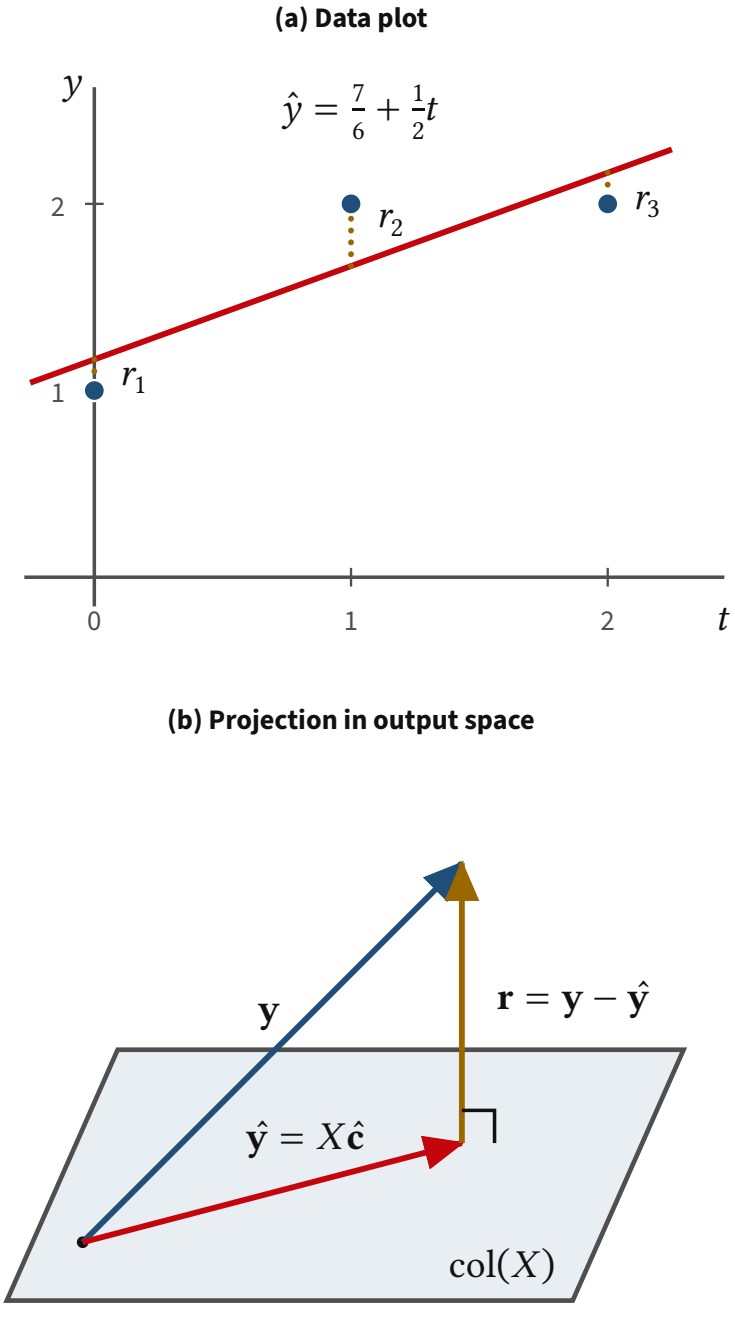


Figure. Two vertically arranged panels show the same linear regression. The upper panel (a) contains three data points, a fitted line, and vertical residual segments. The lower panel (b) shows the data vector y projected onto the column space of X, with a perpendicular residual r.

The two panels in the figure show different spaces: the upper panel (a) is the t-y data plot, while the lower panel (b) is the output space $\mathbb{R}^3$.

Activity: Reading regression code (U4-LO4, U4-LO7)

The following code repeats the line fit from Fit a line to three data points (U4-LO4, U4-LO5). The expression `np.ones_like(t)` creates an array of ones with the same shape and data type as `t`. For the array `t` used below, it produces `array([1., 1., 1.])`. The call `np.column_stack([np.ones_like(t), t])` uses these two arrays as columns of the design matrix: the first column contains ones, and the second contains the input values in `t`. Thus

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

The NumPy function `np.linalg.lstsq(X, y, rcond=None)` returns a tuple containing the least-squares coefficient vector, an array of residual sums of squares, the rank of X, and its singular values (to which we will come back in Eigenvalues and second-order optimization). We use `[0]` to select the first item, the coefficient vector. The parameter `rcond` controls the relative numerical cutoff used to determine the matrix rank; `rcond=None` uses NumPy's default cutoff.

```
import numpy as np

t = np.array([0.0, 1.0, 2.0])
y = np.array([1.0, 2.0, 2.0])

X = np.column_stack([np.ones_like(t), t])

c = np.linalg.lstsq(X, y, rcond=None)[0]
yhat = X @ c
r = y - yhat
```

Activity: Reading regression code (U4-LO4, U4-LO7) (continued)

```
X, c, yhat, r, X.T @ r, np.linalg.norm(r)**2
```

1. What is the shape of `X`?
2. Why does `c` have two entries?
3. What mathematical objects are stored in `yhat` and `r`?
4. Why is the residual check `X.T @ r` rather than `X @ r`?
5. What should `X.T @ r` be close to?
6. Does a nonzero residual vector mean the code failed?
7. What quantity is computed by `np.linalg.norm(r)**2`?

Explore least squares interactively with the Residual Squares mini-game.